

Binomial Coefficients and the Structure of an Expansion

Expanding binomial powers, picking out a named term, using the structure of Pascal's triangle, and summing coefficients

- In $(A + B)^n$ written in descending powers of A , the term at index k is $\binom{n}{k}A^{n-k}B^k$, and k starts at 0.
 - Not every problem wants a full expansion. Read the question first: sometimes one term, one coefficient, or one total is all that is asked for.
 - Leave answers exact, and keep signs attached to the number they belong to.
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1. Expand $(x + 1)^3$ completely.

Answer: _____

2. Expand $(x - 4)^3$ completely.

Answer: _____

3. Find the coefficient of x^3 in the expansion of $(x + 2)^5$. Do not expand the whole thing.

Answer: _____

4. Find the sum of all the coefficients in the expansion of $(2x + 3)^4$, without expanding it.

Answer: _____

5. Evaluate $\sum_{k=0}^5 \binom{5}{k}$, the total of row 5 of Pascal's triangle.

Answer: _____

6. Expand $(2x - 1)^4$ completely.

Answer: _____

7. Find the coefficient of x^2 in the expansion of $(3x - 2)^5$.

Answer: _____

8. Every row of Pascal's triangle reads the same forwards and backwards, so binomial coefficients come in mirror-image pairs.

(a) Compute $\binom{9}{3}$.

Answer: _____

(b) Explain why $\binom{9}{3}$ and $\binom{9}{6}$ must be equal, without computing the second one.

Answer: _____

- 9.** For each whole number n , the coefficients of $(1+x)^n$ add up to 2^n . Evaluate $\sum_{n=0}^5 2^n$, the total of those sums for $n = 0$ through $n = 5$.

Answer: _____

- 10.** Find the coefficient of x^4y^2 in the expansion of $(x+y)^6$.

Answer: _____

- 11.** Expand $(x^2+3)^3$ completely, in descending powers of x .

Answer: _____

12. The expansion of $(1 + x)^4$ is the key to all three parts.

(a) Expand $(1 + x)^4$ completely.

Answer: _____

(b) Use your expansion to evaluate 3^4 by substituting a suitable value of x .

Answer: _____

(c) For some whole number n , the coefficients of $(1 + x)^n$ add up to 512. Find n , and explain why substituting $x = 1$ gives the sum of the coefficients.

Answer: _____